1 Question:

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 11 \\ 0 & 1 & 0 & -3 & -1 & 0 \\ 2 & 0 & 5 & 0 & -4 & 27 \\ 3 & 2 & 7 & -6 & -8 & 38 \end{array}\right] \xrightarrow[-3R_1+R_4\to R_4]{-2R_1+R_3\to R_3} \left[\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 11 \\ 0 & 1 & 0 & -3 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 5 \\ 0 & 2 & 1 & -6 & -2 & 5 \end{array}\right]$$

$$\xrightarrow{-2R_2+R_4\to R_4} \left[\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 11 \\ 0 & 1 & 0 & -3 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 0 & 5 \end{array}\right]$$

$x_1 = 1 + 2b$

$x_2 = 3a + b$

$x_3 = 5$

$x_4 = a$

$x_5 = b$

$a, b \in R$

$$\xrightarrow{-2R_3+R_1\to R_1} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -2 & 1 \\ 0 & 1 & 0 & -3 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 0 & 5 \end{array}\right]$$

* Infinitely many sulubion

$a=1$, $b=1$	$a=2$ $b=2$
$x_1 = 3$	$x_1 = 5$
$x_2 = 4$	$x_2 = 8$
$x_3 = 5$	$x_3 = 5$
$x_4 = 1$	$x_4 = 2$
$x_5 = 1$	$x_5 = 2$

Question 2

$$\left[\begin{array}{cccc|c} 1 & 1 & c & 1 & c \\ 0 & -1 & 1 & 2 & 0 \\ 1 & 2 & 1 & -1 & -c \end{array}\right] \xrightarrow{-R_1+R_3} \left[\begin{array}{cccc|c} 1 & 1 & c & 1 & c \\ 0 & -1 & 1 & 2 & 0 \\ 0 & 1 & 1-c & -2 & -2c \end{array}\right]$$

$$\xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{cccc|c} 1 & 1 & c & 1 & c \\ 0 & -1 & 1 & 2 & 0 \\ 0 & 0 & 2-c & 0 & -2c \end{array}\right] \xrightarrow[\frac{1}{2-c}R_3\to R_3]{-R_2\to R_2} \left[\begin{array}{cccc|c} 1 & 1 & c & 1 & c \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 1 & 0 & \frac{-2c}{2-c} \end{array}\right]$$

c) $c=2 \Rightarrow$ no solution $\quad \frac{-2.2}{2-2} = \frac{-4}{0}$

b) $ç.k = \{\emptyset\}$

a) $ç.k = R - \{2\}$

4 Que

3 Question

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 3 & -1 & 5 & 2 \\ 4 & 1 & (a^2-14) & a+2 \end{array}\right] \xrightarrow[-4R_1+R_3]{-3R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & -7 & a^2-2 & a-14 \end{array}\right]$$

$$\xrightarrow{-R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & 0 & a^2-16 & a-4 \end{array}\right] \xrightarrow{-\frac{1}{7}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & 1 & -2 & 10/7 \\ 0 & 0 & a^2-16 & a-4 \end{array}\right]$$

no solution $\Rightarrow$ Ç.K $= \{-4\}$ $\qquad 0 \quad 0 \quad 0 \;|\; 8$

infinite many solution $\Rightarrow$ Ç.K $= \{4\}$

Exactly one solution $\Rightarrow$ Ç.K $= \mathbb{R} - \{-4, 4\}$

4 Question

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \\ -2 & 1 & 1 \end{bmatrix} \xrightarrow[E_1]{R_1 + R_3 \to R_3} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 1 \end{bmatrix} \xrightarrow[E_2]{-2R_2 + R_3 \to R_3} \underbrace{\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \end{bmatrix}}_{U}$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \qquad E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \qquad E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

$\times$ $L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$

$E_2 \, E_1 \, A = U$

$A = \underbrace{E_1^{-1} \cdot E_2^{-1}}_{L} \cdot U$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

$y_1 = 1 \quad y_2 = 2 \quad y_3 = -4 \qquad -y_1 + 2y_2 + y_3 = -2$

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix}$$

$x_3 = \frac{-4}{3} \quad x_2 = \frac{2}{3} \quad x_1 = \frac{1}{6}$

$2x_1 + x_2 = 1$

$x_2 - x_3 = 2$

5

5 Question

$$A \to \left[\begin{array}{cccc|cccc} 2 & 0 & -1 & 3 & 1 & 0 & 0 & 0 \\ 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 4 & 1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 1 & 3 & -2 & -5 & 0 & 0 & 0 & 1 \end{array}\right]$$

$-R_2 + R_4 \to R_4$

$-2R_2 + R_1 \to R_1$

$-4R_2 + R_3 \to R_3$

$$\left[\begin{array}{cccc|cccc} 0 & 4 & -7 & 1 & 1 & -2 & 0 & 0 \\ 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 0 & 9 & -12 & -5 & 0 & -4 & 1 & 0 \\ 0 & 5 & -5 & -6 & 0 & -1 & 0 & 1 \end{array}\right]$$

$R_2 \leftrightarrow R_1$

$$\left[\begin{array}{cccc|cccc} 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 0 & 4 & -7 & 1 & 1 & -2 & 0 & 0 \\ 0 & 9 & -12 & -5 & 0 & -4 & 1 & 0 \\ 0 & 5 & -5 & -6 & 0 & -1 & 0 & 1 \end{array}\right]$$

$-\frac{9}{4} R_2 + R_3 \to R_3$

$-\frac{5}{4} R_2 + R_3 \to R_3$

$$\left[\begin{array}{cccc|cccc} 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 0 & 4 & -7 & 1 & 1 & -2 & 0 & 0 \\ 0 & 0 & 15/4 & -29/4 & -\frac{9}{4} & \frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{15}{4} & -\frac{29}{4} & -\frac{5}{4} & \frac{6}{4} & 0 & 1 \end{array}\right]$$

$-R_3 + R_4 \to R_4$

$$\left[\begin{array}{cccc|cccc} 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 0 & 4 & -7 & 1 & 1 & -2 & 0 & 0 \\ 0 & 0 & 15/4 & -\frac{29}{4} & -\frac{9}{4} & \frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right]$$

Matrix A has no inverse

5/

$$B \rightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & -3 & 0 & 1 & 0 & 0 \\ 0 & 2 & -3 & 7 & 0 & 0 & 1 & 0 \\ 2 & 1 & -2 & 12 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[-2R_1 + R_4 \to R_4]{R_1 + R_2 \to R_2}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 2 & -3 & 7 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & -2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[-R_2 + R_4 \to R_4]{-2R_2 + R_3 \to R_3}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -2 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & -3 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_3 + R_4 \to R_4}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -2 & -2 & 1 & 0 \\ 0 & 0 & 0 & 3 & -5 & -3 & 1 & 1 \end{array}\right] \xrightarrow[\frac{1}{3} R_4 \to R_4]{-R_3 \to R_3}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -3 & 2 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 & -\frac{5}{3} & -1 & \frac{1}{3} & \frac{1}{3} \end{array}\right] \begin{array}{c} -5R_4 + R_1 \to R_1 \\ -3R_4 + R_3 \to R_3 \\ \xrightarrow[-2R_4 + R_2 \to R_2]{\hspace{3cm}} \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 0 & 28/3 & 5 & -5/3 & -5/3 \\ 0 & 1 & -1 & 0 & 13/3 & 3 & -2/3 & -2/3 \\ 0 & 0 & 1 & 0 & -3 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -5/3 & -1 & 1/3 & 1/3 \end{array}\right] \xrightarrow[R_3 + R_1 \to R_1]{R_3 + R_2 \to R_2}$$

$$\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \underbrace{\left[\begin{array}{cccc} 19/3 & 4 & -5/3 & -2/3 \\ 4/3 & 2 & -2/3 & 1/3 \\ -3 & -1 & 0 & 1 \\ -5/3 & -1 & 1/3 & 1/3 \end{array}\right]}_{B^{-1}}$$

6. Question

$$A=\begin{bmatrix}1&0&-1\\0&6&-1\\0&0&4\end{bmatrix}\underbrace{\overbrace{\xrightarrow[\frac{1}{4}R_3\to R_3]{\frac{1}{6}R_2\to R_2}}^{E1}}_{E_2}\begin{bmatrix}1&0&-1\\0&1&-1/6\\0&0&1\end{bmatrix}\underbrace{\overbrace{\xrightarrow[\frac{1}{6}R_3+R_2\to R_2]{R_3+R_1\to R_1}}^{E_3}}_{E_4}\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$$E_1=\begin{bmatrix}1&0&0\\0&1/6&0\\0&0&1\end{bmatrix}\quad E_2=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1/4\end{bmatrix}$$

$$E_3=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\xrightarrow{R_3+R_1\to R_1}\begin{bmatrix}1&0&1\\0&1&0\\0&0&1\end{bmatrix}$$

$$A^{-1}=E_4.E_3.E_2.E_1.I$$

$$E_4=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\xrightarrow{\frac{1}{6}R_3+R_2\to R_2}\begin{bmatrix}1&0&0\\0&1&1/6\\0&0&1\end{bmatrix}$$

$$B=\begin{bmatrix}1&0&-2\\0&2&1\\0&0&1\end{bmatrix}\xrightarrow[E_1]{\frac{1}{2}R_2\to R_2}\begin{bmatrix}1&0&-2\\0&1&1/2\\0&0&1\end{bmatrix}\underbrace{\overbrace{\xrightarrow[-\frac{1}{2}R_3+R_2\to R_2]{2R_3+R_1\to R_1}}^{E_2}}_{E_3}\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$$E_1=\begin{bmatrix}1&0&0\\0&1/2&0\\0&0&1\end{bmatrix}$$

$$E_2=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\xrightarrow{2R_3+R_1\to R_1}\begin{bmatrix}1&0&2\\0&1&0\\0&0&1\end{bmatrix}$$

$$B^{-1}=E_3.E_2.E_1 I$$

$$E_3=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\xrightarrow{-\frac{1}{2}R_3+R_2\to R_2}\begin{bmatrix}1&0&0\\0&1&-1/2\\0&0&1\end{bmatrix}$$

7 Question

$$\begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & 0 & c \end{bmatrix}$$

$$\begin{bmatrix} ab+1 & a & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$\begin{bmatrix} ab+1 & a & 0 \\ b & 1 & 0 \\ 0 & 0 & c \end{bmatrix} \xrightarrow[\frac{1}{c} R_3 \to R_3]{-aR_2 + R_1 \to R_1} \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{-bR_1 + R_2 \to R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[\frac{1}{c} R_3 \to R_c]{-aR_2 + R_1 \to R_1} \begin{bmatrix} 1 & -a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/c \end{bmatrix}$$

$$\xrightarrow{-bR_1 + R_2 \to R_2} \begin{bmatrix} 1 & -a & 0 \\ -b & ab+1 & 0 \\ 0 & 0 & \frac{1}{c} \end{bmatrix} = A^{-1}$$

8
Question

$(1,4)$ $a_2 + a_1 + a_0 = 4$

$(2\ 0)$ $4a_2 + 2a_1 + a_0 = 0$

$(3\ 12)$ $9a_2 + 3a_1 + a_0 = 12$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 1 & 2 & 4 & 0 \\ 1 & 3 & 9 & 12 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & -4 \\ 0 & 2 & 8 & 8 \end{array}\right]$$

$$\xrightarrow{-2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & -4 \\ 0 & 0 & 2 & 10 \end{array}\right]$$

$a_2 = 8$ $a_1 = -28$ $a_0 = 24$

$a_1 + 3\overset{24}{\widehat{a_3}} = -4$

$a_0 + a_1 + a_2 = 4$

CamScanner ile tara

9 Question

$$
\begin{aligned}
x_1 - x_2 \qquad\qquad\qquad\qquad &= 400 \\
x_1 \qquad + x_3 - x_4 \qquad &= 600 \\
x_2 + x_3 \qquad + x_5 &= 300 \\
x_4 + x_5 &= 100
\end{aligned}
$$

$$
\left[\begin{array}{ccccc|c}
1 & -1 & 0 & 0 & 0 & 400 \\
1 & 0 & 1 & -1 & 0 & 600 \\
0 & 1 & 1 & 0 & 1 & 300 \\
0 & 0 & 0 & 1 & 1 & 100
\end{array}\right]
\xrightarrow{-R_1 + R_2 \to R_2}
$$

$$
\left[\begin{array}{ccccc|c}
1 & -1 & 0 & 0 & 0 & 400 \\
0 & 1 & 1 & -1 & 0 & 200 \\
0 & 1 & 1 & 0 & 1 & 300 \\
0 & 0 & 0 & 1 & 1 & 100
\end{array}\right]
\xrightarrow{-R_2 + R_3 \to R_3}
$$

$$
\left[\begin{array}{ccccc|c}
1 & -1 & 0 & 0 & 0 & 400 \\
0 & 1 & 1 & -1 & 0 & 200 \\
0 & 0 & 0 & 1 & 1 & 100 \\
0 & 0 & 0 & 1 & 1 & 100
\end{array}\right]
\xrightarrow{-R_3 + R_4 \to R_4}
$$

$$
\left[\begin{array}{ccccc|c}
1 & -1 & 0 & 0 & 0 & 400 \\
0 & 1 & 1 & -1 & 0 & 200 \\
0 & 0 & 0 & 1 & 1 & 100 \\
0 & 0 & 0 & 0 & 0 & 0
\end{array}\right]
$$

$x_1 = -300$

$x_2 = -700$

$x_3 = 0$

$x_4 = -900$

$x_5 = 1000$

10 Question

$$
\begin{aligned}
-x_1 + x_2 \quad + x_4 \quad &= 200 \\
x_1 \quad + x_3 \quad - x_6 &= 100 \\
x_2 + x_3 \quad - x_5 \quad &= 300 \\
x_4 + x_5 - x_6 &= 0
\end{aligned}
$$

$$
\left[\begin{array}{cccccc|c}
-1 & 1 & 0 & 1 & 0 & 0 & 200 \\
1 & 0 & 1 & 0 & 0 & -1 & 100 \\
0 & 1 & 1 & 0 & -1 & 0 & 300 \\
0 & 0 & 0 & 1 & 1 & -1 & 0
\end{array}\right]
\xrightarrow{R_1 + R_2 \to R_2}
$$

$$
\to \left[\begin{array}{cccccc|c}
-1 & 1 & 0 & 1 & 0 & 0 & 200 \\
0 & 1 & 1 & 1 & 0 & -1 & 300 \\
0 & 1 & 1 & 0 & -1 & 0 & 300 \\
0 & 0 & 0 & 1 & 1 & -1 & 0
\end{array}\right]
\xrightarrow{-R_2 + R_3 \to R_3}
$$

$$
\to \left[\begin{array}{cccccc|c}
-1 & 1 & 0 & 1 & 0 & 0 & 200 \\
0 & 1 & 1 & 1 & 0 & -1 & 300 \\
0 & 0 & 0 & -1 & -1 & 1 & 0 \\
0 & 0 & 0 & 1 & 1 & -1 & 0
\end{array}\right]
\xrightarrow{R_3 + R_4 \to R_4}
$$

$$
\left[\begin{array}{cccccc|c}
-1 & 1 & 0 & 1 & 0 & 0 & 200 \\
0 & 1 & 1 & 1 & 0 & -1 & 300 \\
0 & 0 & 0 & -1 & -1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0
\end{array}\right]
$$

$x_1 = -850$

$x_2 = -650$

$x_3 = 1000$

$x_4 = 0$

$x_5 = 50$

$x_6 = 50$

11 Question

$$
\begin{aligned}
I_1 - I_2 + I_3 &= 0 \\
I_1 - I_2 + I_4 &= 0 \\
I_3 - I_5 + I_6 &= 0 \\
I_4 - I_5 + I_6 &= 0 \\
3I_1 + 2I_2 &= 14 \\
2I_2 + 4I_3 + 2I_4 + I_5 &= 25 \\
I_5 + I_6 &= 8
\end{aligned}
$$

1 2

3 4

$$
\left[\begin{array}{cccccc|c}
1 & -1 & 1 & 0 & 0 & 0 & 0 \\
1 & -1 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & -1 & 1 & 0 \\
0 & 0 & 0 & 1 & -1 & 1 & 0 \\
3 & 2 & 0 & 0 & 0 & 0 & 14 \\
0 & 2 & 4 & 2 & 1 & 0 & 25 \\
0 & 0 & 0 & 0 & 1 & 1 & 8
\end{array}\right]
\xrightarrow[-3R_1 + R_5 \to R_5]{-R_1 + R_2 \to R_2}
\left[\begin{array}{cccccc|c}
1 & -1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & -1 & 1 & 0 \\
0 & 0 & 0 & 1 & -1 & 1 & 0 \\
0 & 5 & -3 & 0 & 0 & 0 & 14 \\
0 & 2 & 4 & 2 & 1 & 0 & 25 \\
0 & 0 & 0 & 0 & 1 & 1 & 8
\end{array}\right]
$$

$$
\xrightarrow[R_2 \leftrightarrow R_5]{-\frac{2}{5}R_5 + R_6 \to R_6}
\left[\begin{array}{cccccc|c}
1 & -1 & 1 & 0 & 0 & 0 & 0 \\
0 & 5 & -3 & 0 & 0 & 0 & 14 \\
0 & 0 & 1 & 0 & -1 & 1 & 0 \\
0 & 0 & 0 & 1 & -1 & 1 & 0 \\
0 & 0 & -1 & 1 & 0 & 0 & 0 \\
0 & 0 & \frac{26}{5} & 2 & 1 & 0 & \frac{97}{5} \\
0 & 0 & 0 & 0 & 1 & 1 & 8
\end{array}\right]
$$

$$
\xrightarrow[-\frac{26}{5}R_3 + R_6 \to R_6]{R_3 + R_5 \to R_5}
\left[\begin{array}{cccccc|c}
1 & -1 & 1 & 0 & 0 & 0 & 0 \\
0 & 5 & -3 & 0 & 0 & 0 & 14 \\
0 & 0 & 1 & 0 & -1 & 1 & 0 \\
0 & 0 & 0 & 1 & -1 & 1 & 0 \\
0 & 0 & 0 & 1 & -1 & 1 & 0 \\
0 & 0 & 0 & 2 & \frac{31}{5} & \frac{26}{5} & \frac{97}{5} \\
0 & 0 & 0 & 0 & 1 & 1 & 8
\end{array}\right]
$$

11

$$\xrightarrow[\substack{-2R_4+R_6\to R_6 \\ \frac{1}{5}R_2\to R_2}]{-R_4+R5\to R_5}\left[\begin{array}{cccccc|c} 1 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -3/5 & 0 & 0 & 0 & 14/5 \\ 0 & 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{41}{5} & \frac{36}{5} & \frac{97}{5} \\ 0 & 0 & 0 & 0 & 1 & 1 & 8 \end{array}\right]$$

$$\xrightarrow[R_7\leftrightarrow R_5]{-\frac{41}{5}R_7+R_6\to R_6}\left[\begin{array}{cccccc|c} 1 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -3/5 & 0 & 0 & 0 & 14/5 \\ 0 & 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 8 \\ 0 & 0 & 0 & 0 & 0 & -1 & \frac{-231}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$I_6 = \frac{231}{5}$ amps

$I_5 = \frac{-191}{5}$ amps

$I_4 = \frac{-422}{5}$ amps

$I_3 = \frac{-422}{5}$ amps

$I_2 = \frac{-1194}{25}$ amps

$I_1 = \frac{-916}{25}$ amps